

☒ Easy Level (10 Questions)

1.

Which command is used to schedule a task in Linux?

- A) cron
- B) auto
- C) bashd
- D) systask

Answer: A

☒ cron is the standard scheduler in Unix-like systems.

2.

What does the following command do?

```
bash
```

Copy code

```
tar -czf backup.tar.gz /home/user
```

- A) Deletes files
- B) Compresses and archives
- C) Lists contents
- D) Moves files

Answer: B

3.

Which tool is used in Ubuntu to update system packages?

- A) rpm
- B) dnf
- C) apt
- D) patch

Answer: C

4.

What is the purpose of unattended-upgrades?

- A) Boot loader setup
- B) Install security patches automatically
- C) Reset user permissions
- D) Clean unused files

Answer: B

5.

What will this script do?

bash

Copy code

```
find /tmp -type f -mtime +3 -delete
```

- A) Archive files
- B) Rename files
- C) Delete files older than 3 days
- D) Move files

Answer: C

6.

Where do you define cron jobs for the current user?

- A) /etc/bash.bashrc
- B) crontab -e

C) /root/cronjobs

D) /etc/sudoers

Answer: B

7.

What does apt-get upgrade do?

A) Installs new packages

B) Upgrades installed packages

C) Removes packages

D) Shows patch report

Answer: B

8.

Which command will apply all available security updates on RHEL?

A) yum install all

B) dnf patch

C) yum update -y

D) apt install patch

Answer: C

9.

What is the default logging tool in most Linux systems?

- A) logger
- B) syslog
- C) netlog
- D) errorctl

Answer: B

10.

How often can a cron job run?

- A) Only hourly
- B) Only daily
- C) Once per week
- D) As frequently as every minute

Answer: D

☒ Medium Level (20 Questions)

11.

Which field in a crontab line specifies the hour?

- A) 1st

- B) 2nd
- C) 3rd
- D) 5th

Answer: B

⌘ Format: min hour dom month dow command

12.

What does the 2>&1 do in a command like this?

bash

Copy code

```
script.sh > output.txt 2>&1
```

- A) Disables script
- B) Merges stderr into stdout
- C) Duplicates output
- D) Deletes errors

Answer: B

13.

Which tool on Debian automates security updates?

- A) yum
- B) unattended-upgrades
- C) dpkg-patcher
- D) patch-it

Answer: B

14.

What does set -e ensure in a script?

- A) Debug logging
- B) Script skips errors
- C) Script exits on error
- D) Executes silently

Answer: C

15.

How would you clean up .log files older than 30 days?

bash

Copy code

```
? /var/log -name "*.log" -mtime +30 -exec rm {} \;
```

- A) del
- B) grep
- C) find
- D) awk

Answer: C

16.

Which command upgrades packages in RHEL using dnf?

- A) dnf install update
- B) dnf update -y
- C) dnf upgrade
- D) dnf refresh

Answer: B

17.

What does this cron line mean?

swift

Copy code

```
0 3 * * 1 /usr/local/bin/cleanup.sh
```

- A) Runs at 3 PM daily
- B) Runs every minute
- C) Runs every Monday at 3 AM
- D) Runs at random time

Answer: C

18.

Which of these methods is most reliable for large-scale automated patching?

- A) apt-get
- B) Manual install
- C) Ansible

D) kill -HUP

Answer: C

19.

What is the purpose of /dev/null in automation scripts?

A) Deletes logs

B) Stores backups

C) Discards unwanted output

D) Sends mail

Answer: C

20.

Which symbol in a script denotes a comment?

A) //

B) ;

C) #

D) \$

Answer: C

21.

How to write a Bash loop to apply patches on multiple servers?

- A) if loop
- B) select loop
- C) for loop
- D) while case

Answer: C

22.

What does the command yum clean all do?

- A) Deletes all packages
- B) Cleans cache metadata
- C) Uninstalls yum
- D) Resets the repo

Answer: B

23.

Which script pattern should be used before running risky commands?

- A) set +u
- B) trap ERR
- C) echo "Hi"
- D) break

Answer: B

❑ trap is used to catch errors.

24.

Which log file shows yum operations?

- A) /var/log/secure
- B) /etc/yum.log
- C) /var/log/yum.log
- D) /var/log/messages

Answer: C

25.

Why is chmod +x script.sh necessary?

- A) Makes file read-only
- B) Enables patching
- C) Grants execution permission
- D) Edits file

Answer: C

26.

Which command reads crontab entries for a user?

- A) cat /etc/crontab
- B) ls cron
- C) crontab -l
- D) grep cron

Answer: C

27.

In security patching, what is “dependency hell”?

- A) Too many open ports
- B) Missing patch logs
- C) Conflicts between packages
- D) Server crashes

Answer: C

28.

Which tool helps version control scripts and automation?

- A) zip
- B) rsync
- C) git
- D) ssh

Answer: C

29.

Which package type does rpm manage?

- A) .deb
- B) .zip
- C) .pkg
- D) .rpm

Answer: D

30.

Which command updates only security patches using yum?

- A) yum upgrade
- B) yum update --security
- C) yum secure
- D) yum patch-only

Answer: B

☒ Hard Level (10 Questions)

31.

What does trap 'echo "error"; exit' ERR do?

- A) Ignores errors
- B) Executes code on error
- C) Always runs
- D) Exits on success

Answer: B

32.

Which feature should a patch automation script include?

- A) Shutdown logic
- B) Logging and alerts
- C) User prompts
- D) Hard-coded paths

Answer: B

33.

What does this script do?

bash

Copy code

```
#!/bin/bash for host in server1 server2; do ssh $host 'yum update -y' done
```

- A) Reboots servers
- B) Removes yum
- C) Patches servers remotely
- D) Checks DNS

Answer: C

34.

How to prevent a script from crashing due to undefined variables?

- A) Use set -u
- B) Use set -n
- C) Define all variables
- D) Both A and C

Answer: D

35.

What is a safe way to test a patch automation script?

- A) On production
- B) On staging/test system
- C) On firewall
- D) Without logs

Answer: B

36.

What does cron.d directory contain?

- A) Linux boot logs
- B) Kernel files
- C) System-wide cron jobs
- D) Patches

Answer: C

37.

Which scripting practice increases security in automation?

- A) Plaintext passwords
- B) Logging sudo access
- C) File permissions & logging
- D) Aliases

Answer: C

38.

Which Bash command is best to check for root access?

- A) id -u
- B) whoami
- C) sudo -v
- D) su

Answer: A

☒ If id -u returns 0, user is root.

39.

Which command helps rollback an RPM update?

- A) yum remove
- B) rpm -e
- C) rpm -Uvh
- D) rpm -Uvh --rollback

Answer: D

40.

Best way to alert admin after patching via script?

- A) echo
- B) cat file
- C) sendmail / mailx

D) chmod

Answer: C